

Electronic Mail in the Internet

Introduction

Electronic mail (e-mail) has been one of the most important and widely used applications of the Internet since its inception. Initially the most popular application of the Internet, e-mail has evolved to include powerful features such as:

- Attachments
- Hyperlinks
- HTML-formatted text
- Embedded multimedia (photos, etc.)

E-mail is an **asynchronous communication medium**, allowing users to send and receive messages at their convenience. Compared to traditional postal mail, e-mail is **fast, inexpensive, and easy to distribute**.

Key Components of the Internet Mail System

The Internet mail system consists of three major components (see Figure 1):

1. User Agents (UA)

- Software that allows users to compose, read, forward, reply, and manage e-mails.
- Examples: Microsoft Outlook, Apple Mail, Gmail (web and app).

2. Mail Servers

- Store mailboxes and handle message queues.
- Manage sending, receiving, and retrying delivery of e-mails.
- Authenticate users via credentials.

3. Simple Mail Transfer Protocol (SMTP)

- The core application-layer protocol for e-mail transfer.
- Uses TCP for reliable message delivery.
- Operates in a client-server model: sending server = SMTP client; receiving server = SMTP server.

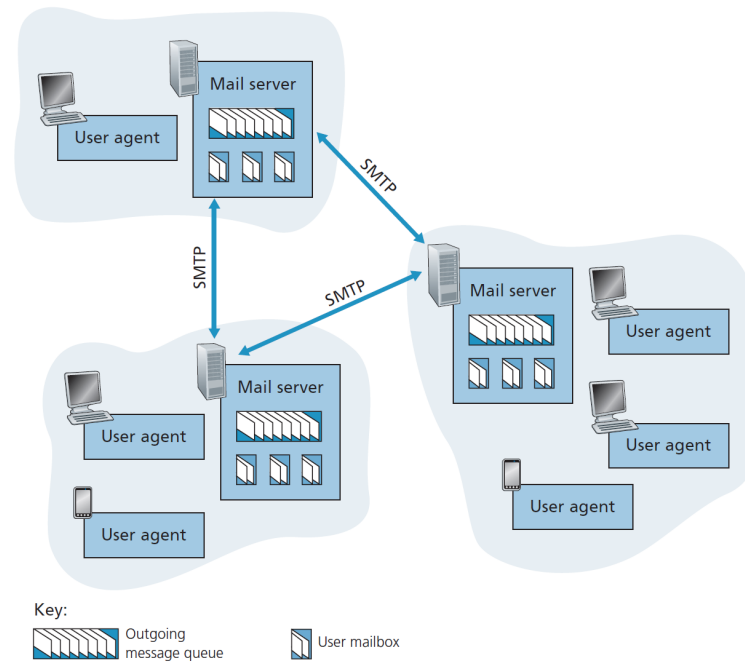


Figure 1: High-level view of the Internet e-mail system.

SMTP (Simple Mail Transfer Protocol)

Defined in **RFC 5321**, SMTP is the protocol responsible for transferring messages between mail servers.

Characteristics

- Uses TCP (port 25) for reliable transmission.
- Employs a client-server model.
- Uses persistent connections to send multiple messages.
- Originally restricted to 7-bit ASCII, requiring multimedia encoding (e.g., Base64).

SMTP Working Example

Consider Alice sending an ASCII message to Bob:

1. Alice composes a message via her **user agent**.
2. The message is sent to **Alice's mail server** and queued.
3. SMTP client (Alice's server) opens a TCP connection to SMTP server (Bob's server).
4. After **handshaking**, Alice's server sends the message.
5. Bob's mail server stores the message in his mailbox.
6. Bob retrieves the message using his **user agent**.

SMTP Command Example

A dialogue between client (C) and server (S):

SMTP Dialogue

```
S: 220 hamburger.edu
C: HELO crepes.fr
S: 250 Hello crepes.fr, pleased to meet you
C: MAIL FROM: <alice@crepes.fr>
S: 250 Sender OK
C: RCPT TO: <bob@hamburger.edu>
S: 250 Recipient OK
C: DATA
S: 354 Enter mail, end with "." on a line by itself
C: Do you like ketchup?
C: How about pickles?
C: .
S: 250 Message accepted
C: QUIT
S: 221 Connection closed
```

Mail Message Formats

Defined in **RFC 5322**, e-mail messages consist of:

- **Header:** Contains metadata.
 - Required: From, To
 - Optional: Subject, Date, CC, etc.
- **Body:** The actual message content (ASCII text).

Example Header:

```
From: alice@crepes.fr
To: bob@hamburger.edu
Subject: Searching for the meaning of life
```

Mail Access Protocols

Once SMTP delivers the message to Bob's mail server, Bob needs to **retrieve** it.

Why Not SMTP?

- SMTP is a **push protocol**, designed to send mail between servers.
- Retrieval requires a **pull protocol**.

Common Retrieval Methods

1. Web-based E-mail (HTTP)

- Example: Gmail, Yahoo Mail.
- User agent retrieves mail via HTTP.
- Supports advanced folder management.

2. IMAP (Internet Mail Access Protocol, RFC 3501)

- Widely used with e-mail clients like Outlook.
- Allows folder creation, message deletion, marking messages, etc.
- Keeps mail on the server, enabling synchronization across devices.

Summary

- E-mail is one of the most essential Internet applications.
- **SMTP** handles sending and transferring messages.
- **Mail servers** store and manage mailboxes.
- **User agents** enable users to compose and manage e-mails.
- **Mail access protocols** like HTTP and IMAP allow retrieval of messages.